

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Generative AI and Large Language Models [NPT_4121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Design an **evaluation framework** to interpret and compare different **word embedding models** (e.g., Word2Vec, GloVe, FastText) using **visualization techniques**. Discuss how visualization helps in diagnosing embedding quality and bias. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Demonstrate how **GPT (Generative Pre-trained Transformer)** functions as a **decoder-only model**. Explain how its architecture enables tasks like text generation and completion. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) **Illustrate** the role of the **Encoder-Decoder model** in **sequence-to-sequence tasks** such as language translation and **show** how this architecture enables a system to convert input sequences into meaningful outputs. (2)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 2) Evaluate the limitations of basic RNN architectures and justify how advanced types like **LSTM** and **GRU** overcome them. Compare their internal mechanisms and performance trade-offs in sequence learning tasks. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply the concept of **ULMFIT** to a sentiment classification task with limited labeled data. Explain how each stage of fine-tuning helps improve model accuracy in this scenario. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) **Apply** your understanding of Artificial Intelligence (AI) to **describe** two key factors that have driven its rapid growth in recent years and **demonstrate** how each factor is reflected in real-world AI applications. (2)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- 3) Analyze the implementation of a **Generative AI system for automated news summarization** using the **Transformer architecture**. Evaluate how each stage of the **Generative AI Project Lifecycle** contributes to ensuring accuracy, fairness, and ethical AI deployment. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Evaluate one **real-world application** of Generative AI and analyze how representation learning enables it to perform complex creative tasks. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Identify the phenomenon of hallucination in Large Language Models (LLMs) and illustrate how it affects the reliability and trustworthiness of AI-generated outputs in real-world applications. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4) **Evaluate why Transformer-based Large Language Models (LLMs)** have become the dominant architecture in modern AI. Justify your answer by analyzing their scalability, efficiency, and emergent capabilities compared to earlier models like RNNs or LSTMs. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Apply the concept of the **Generative AI Project Lifecycle** to describe how a company developing an AI-based chatbot for customer support would structure its workflow from data collection to deployment. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Apply your understanding of Natural Language Processing (NLP) to illustrate any two real-world applications and explain how they create practical impact in solving everyday problems. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 5) **Analyze** the role of **Instruction Tuning** in improving the usability and reliability of large language models. Discuss its effect compared to standard fine-tuning. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply your understanding of **Cost Intuition and Risks** to explain two major challenges faced when deploying LLMs at scale in production environments. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Apply your understanding of Recurrent Neural Network (RNN) architectures to classify the four types of RNNs based on input-output cardinality, and illustrate each type with a suitable real-world example. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4121:Generative AI and Large Language Models

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)